

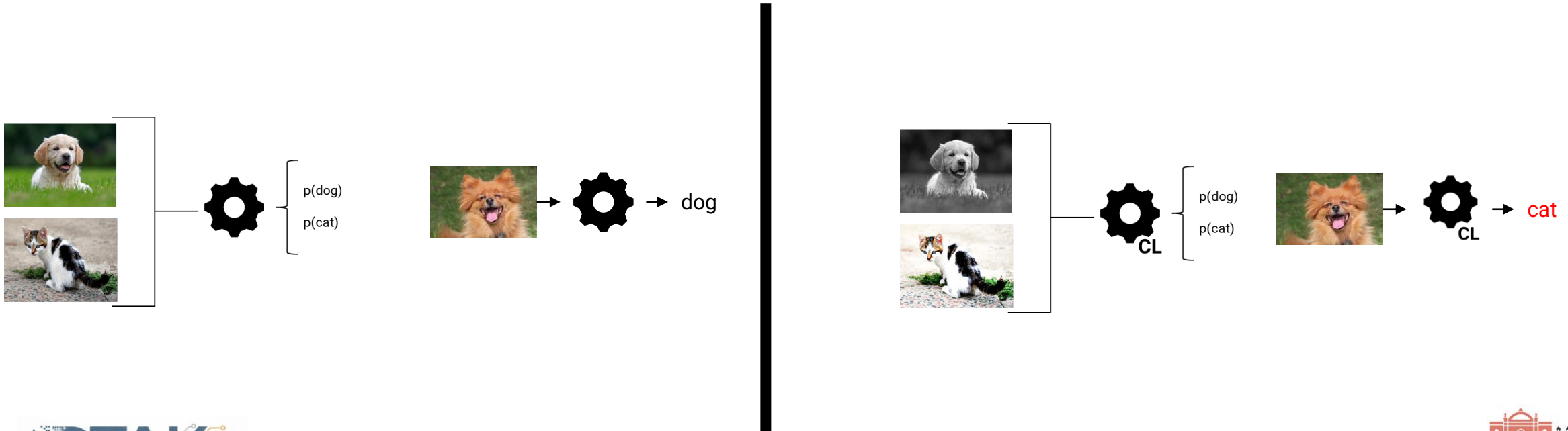
Challenges in CKGE

ISWC'26

Ch 1: Catastrophic Forgetting Characterization

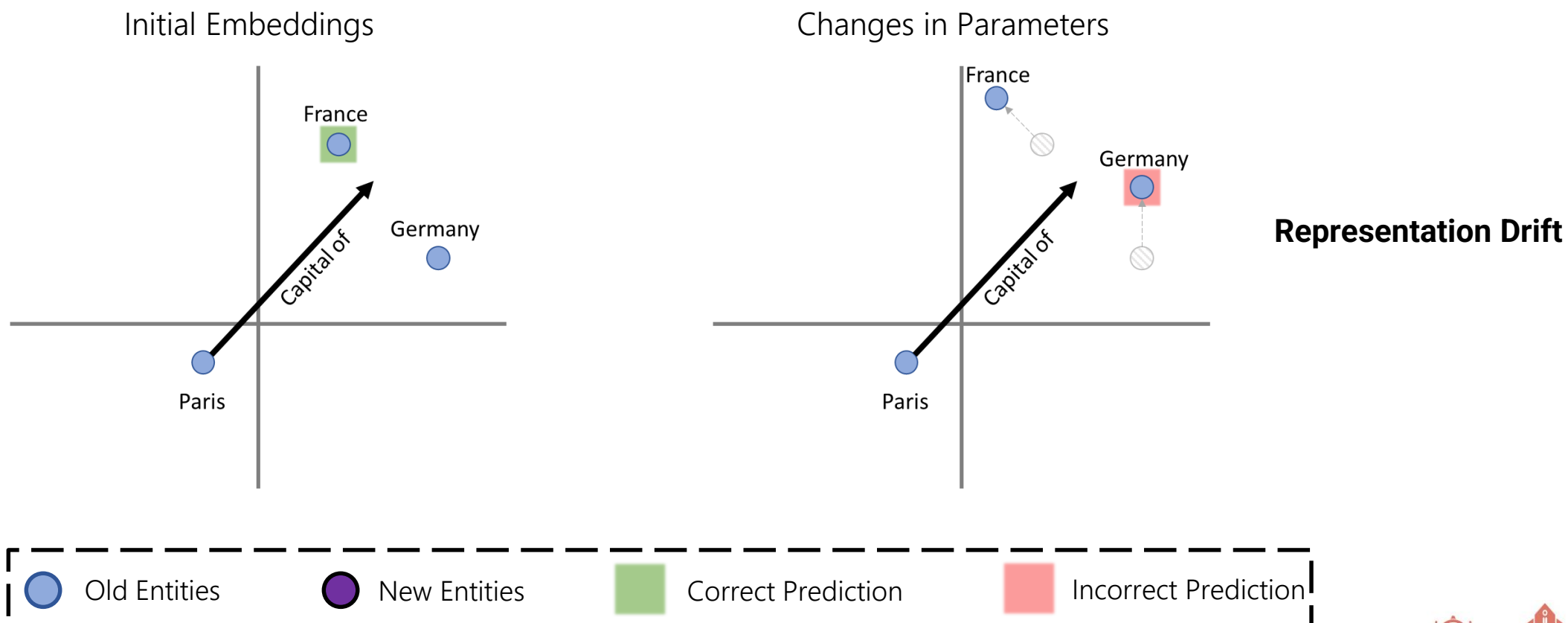
Catastrophic Forgetting in Continual Learning:

In **Continual Learning**, catastrophic forgetting is mostly associated in **undesired changes** in the model parameters (i.e., weights) that **alter the model's output**.



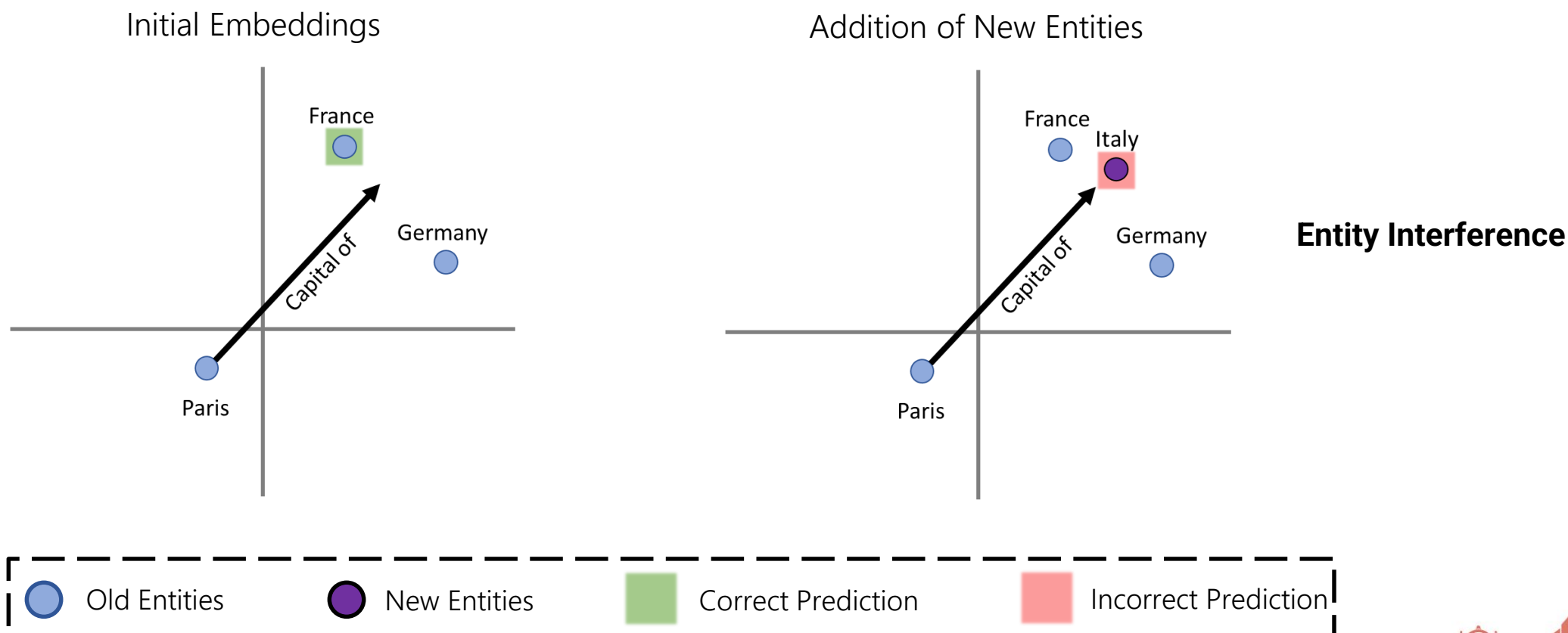
Catastrophic Forgetting in CKGE:

In CKGE, catastrophic forgetting can originate from **two sources**: (Paris, capital of, ?)



Catastrophic Forgetting in CKGE:

In CKGE, catastrophic forgetting can originate from **two sources**: (Paris, capital of, ?)



Catastrophic Forgetting in CKGE:

In CKGE, catastrophic forgetting can originate from **two sources**:

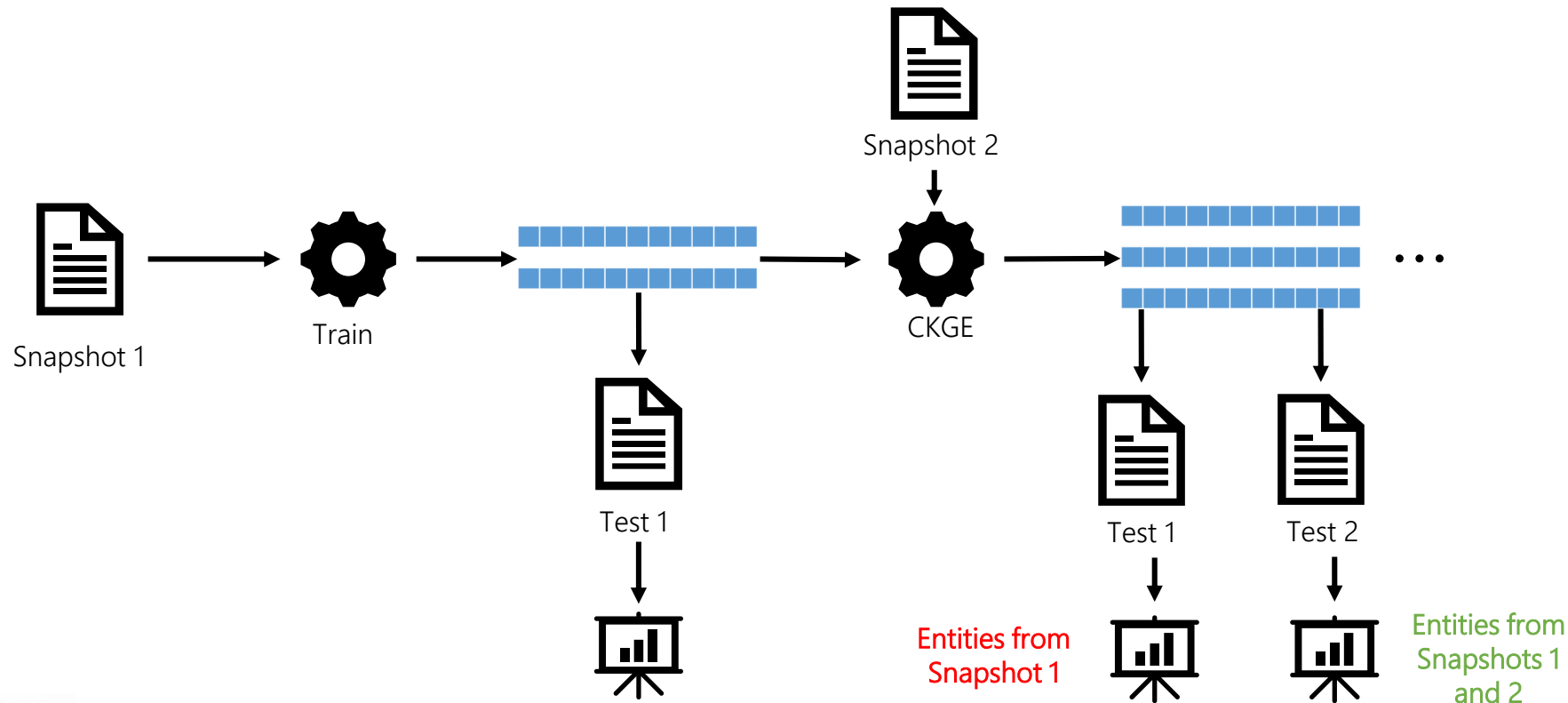
- **Representation Drift**
- **Entity Interference**

However, the latter has been **overlooked by existing CKGE methods**, resulting in:

- Reported **performances** are **overestimated**.
- Catastrophic forgetting is **wrongly assessed**.
- Methods do **not directly tackle entity interference**.

Implication I: Performance Overestimation

Methods that **do not account** for **entity interference** in their evaluation, consider as correct some link prediction results that are in fact **not correct**:

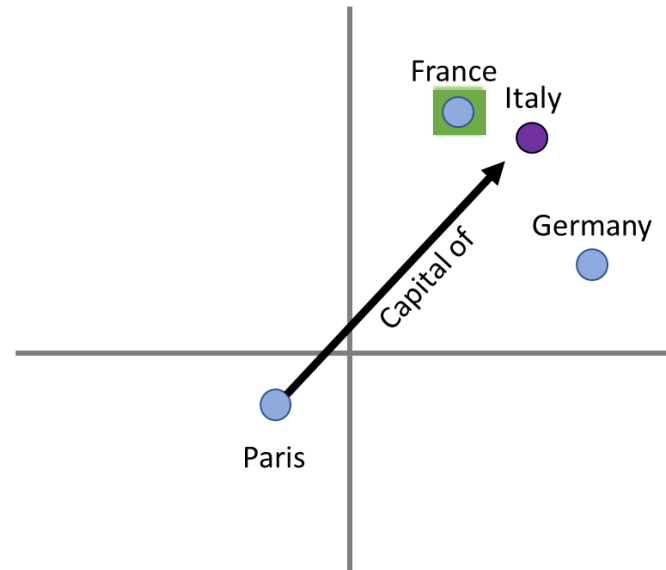


Implication I: Performance Overestimation

Methods that **do not account** for **entity interference** in their evaluation, consider as correct some link prediction results that are in fact **not correct**:

Test 1
(Paris, capital of, ?)

Addition of New Entities



Entities Snapshot 1

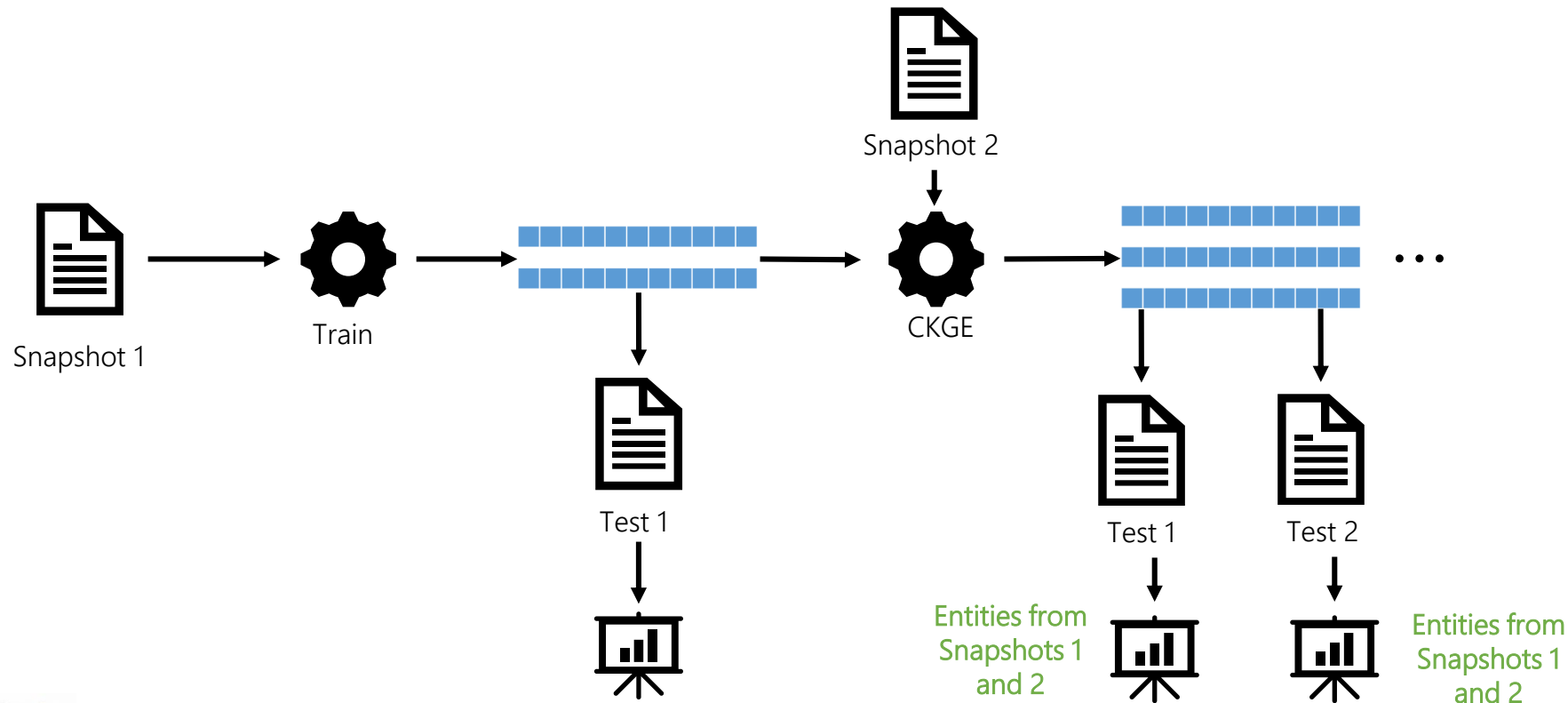
- France
- Germany

Entities Snapshot 2

- Italy

Implication I: Performance Overestimation

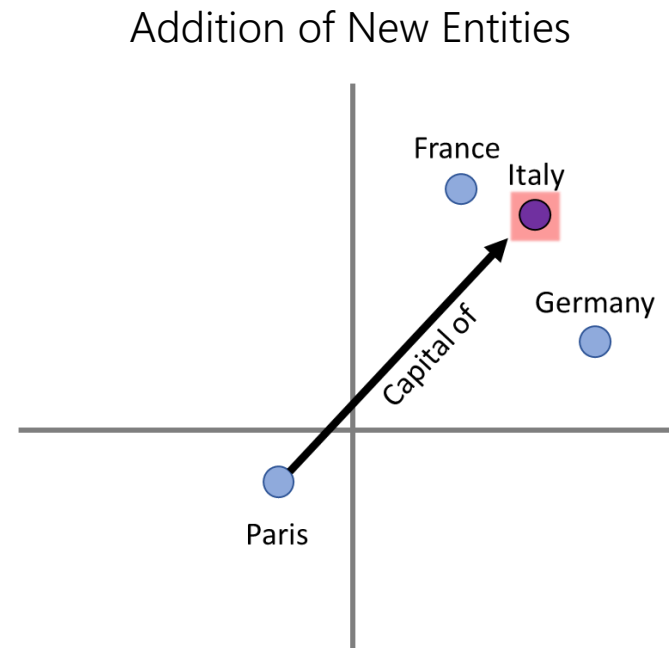
Methods that **do not account** for **entity interference** in their evaluation, consider as correct some link prediction results that are in fact **not correct**:



Implication I: Performance Overestimation

Methods that **do not account** for **entity interference** in their evaluation, consider as correct some link prediction results that are in fact **not correct**:

Test 1
(Paris, capital of, ?)



Implication I: Performance Overestimation

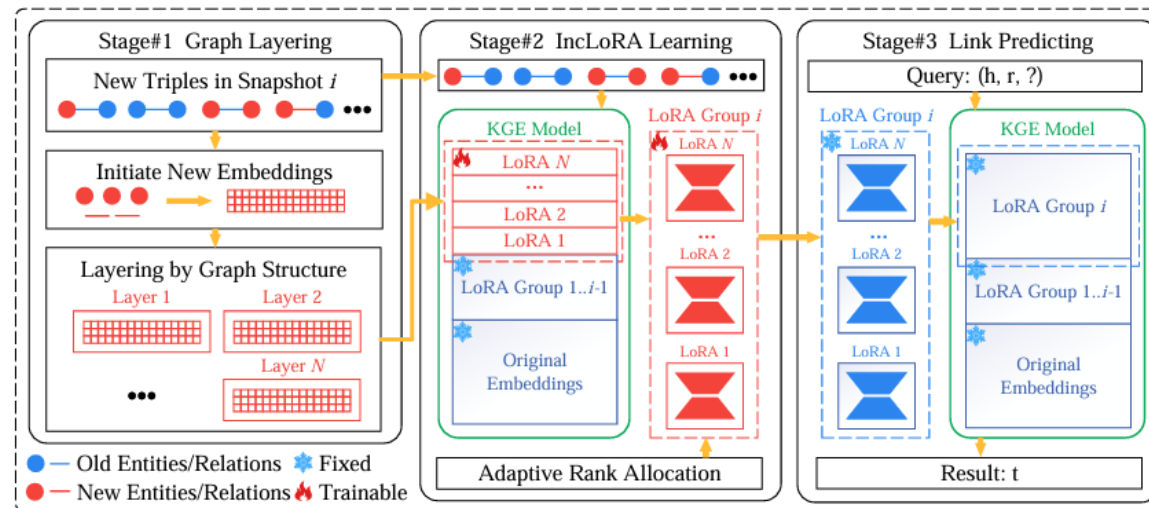
With **state-of-the-art methods** and **benchmark datasets**, this translates to an overestimation of the performance of **up to 25%**.

CKGE	ENTITY		
	$MRR^{w/o}MRR$		%
retraining	0.237	0.211	-11.0
finetune	0.163	0.143	-12.3
EWC	0.228	0.185	-18.9
EMR	0.169	0.148	-12.4
LKGE	0.239	0.185	-22.6
FMR	0.230	0.187	-18.7
ETT-CKGE	0.259	0.200	-22.8
DebiasedKGE	0.237	0.186	-21.5
FastKGE	0.239	0.179	-25.1
SAGE	0.249	0.193	-22.5
incDE	0.253	0.186	-26.5

Implication II: Catastrophic Forgetting

The omission also affects the **interpretation** of catastrophic forgetting. Some methods that were previously seen as totally resistant to catastrophic forgetting are in fact affected by it.

For instance, in FastKGE [1] old embeddings are **frozen**, so they cannot experience **representation drift** → **do not experience forgetting if entity interference is not considered**.



[1] Liu, J., Ke, W., Wang, P., Wang, J., Gao, J., Shang, Z., Li, G., Xu, Z., Ji, K., Li, Y., 2024b. Fast and continual knowledge graph embedding via incremental lora, in: IJCAI.

Implication II: Catastrophic Forgetting

The omission also affects the **interpretation** of catastrophic forgetting. Some methods that were previously seen as totally resistant to catastrophic forgetting are in fact affected by it.

CKGE	ENTITY	RELATION	FACT	HYBRID	GE _{Equal}	GH _{Higher}	GL _{Lower}	PS	Average
finetune	-0.260	-0.220	-0.121	-0.251	-0.103	-0.218	-0.033	-0.116	-0.165
EWC	-0.110	-0.011	-0.024	-0.086	-0.051	-0.212	-0.017	-0.008	-0.065
EMR	-0.245	-0.179	-0.120	-0.232	-0.099	-0.202	-0.045	-0.114	-0.154
LKGE	-0.122	-0.078	-0.054	-0.110	-0.051	-0.149	-0.021	-0.017	-0.075
FMR	-0.101	-0.024	-0.041	-0.098	-0.067	-0.231	-0.030	-0.021	-0.077
ETT-CKGE	-0.133	-0.053	-0.035	-0.065	-0.080	-0.170	-0.045	-0.071	-0.082
DebiasedKGE	-0.113	-0.072	-0.075	-0.115	-0.065	-0.203	-0.047	-0.033	-0.090
→ FastKGE	-0.138	-0.002	-0.011	-0.054	-0.084	-0.202	-0.040	-0.003	-0.067
SAGE	-0.142	-0.098	-0.056	-0.128	-0.081	-0.161	-0.036	-0.028	-0.091
incDE	-0.174	-0.044	-0.025	-0.096	-0.087	-0.180	-0.026	-0.020	-0.082
Average	-0.154	-0.078	-0.056	-0.124	-0.077	-0.193	-0.034	-0.043	—

Implication II: Catastrophic Forgetting

We can also **break down** the contribution of the **two sources of forgetting**:

CKGE	ENTITY		
	$MRR_{0,0}$	$MRR_{N,0}^{w/o}$	$MRR_{N,0}$
finetune	0.303	0.180	0.130
EWC	0.303	0.304	0.192
EMR	0.303	0.190	0.139
LKGE	0.312	0.303	0.175
FMR	0.310	0.310	0.191
ETT-CKGE	0.336	0.326	0.186
DebiasedKGE	0.312	0.310	0.176
FastKGE	0.332	0.332	0.179
SAGE	0.335	0.308	0.170
incDE	0.338	0.308	0.162

Implication II: Catastrophic Forgetting

Or study the effect that catastrophic forgetting has to **different KGE models**.

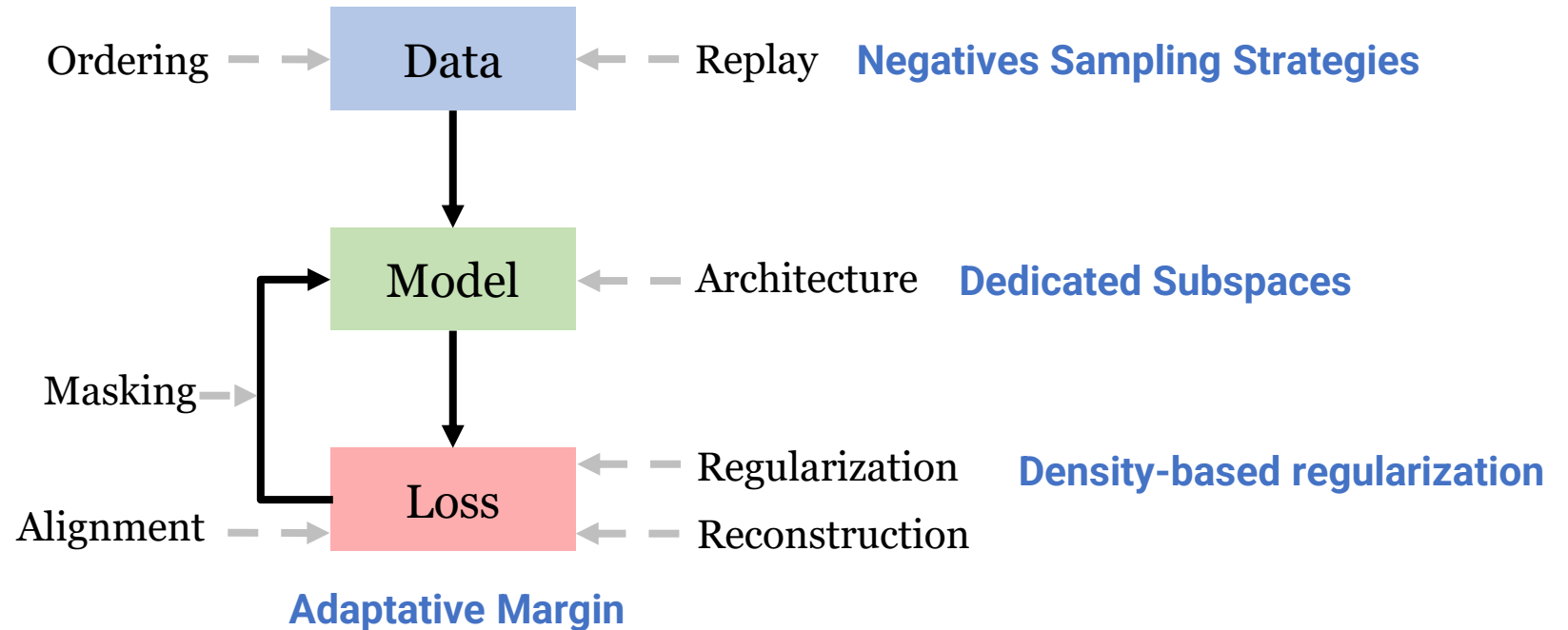
KGE Type	Model	MRR	CF	$MRR_{0,0}$	$MRR_{N,0}^{w/o}$	$MRR_{N,0}$
Translational	TransE	0.226	-0.150	0.339	0.293	0.215
	TransH	0.177	-0.183	0.374	0.209	0.156
	TransR	0.188	-0.028	0.295	0.273	0.270
	RotatE	0.295	-0.192	0.407	0.344	0.223
	TorusE	0.231	-0.197	0.398	0.176	0.072
	BoxE	0.258	-0.234	0.488	0.337	0.205
Semantic	DistMult	0.170	-0.276	0.359	0.210	0.119
	ComplEx	0.112	-0.458	0.354	0.081	0.058
	QuatE	0.159	-0.382	0.466	0.086	0.049
	Simple	0.097	-0.427	0.297	0.097	0.039

Semantic models are the ones **more affected** by both representation drift and entity interference, yet they are **better at knowledge acquisition**.

Translational models, and TransR in particular, show **great stability and robustness to interference**, but they fall behind in knowledge acquisition

Implication III: New Research Lines

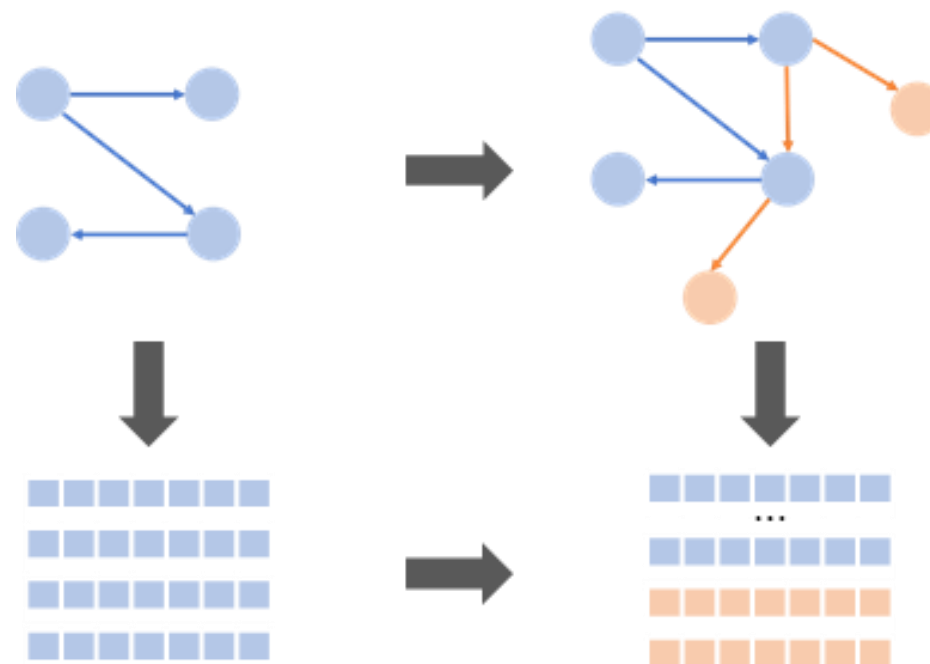
As **entity interference** has been **overlooked**, CKGE methods do not incorporate methods to mitigate it in their architectures. Therefore, **new strategies** could be designed:



Ch 2: Initialization of New Entities

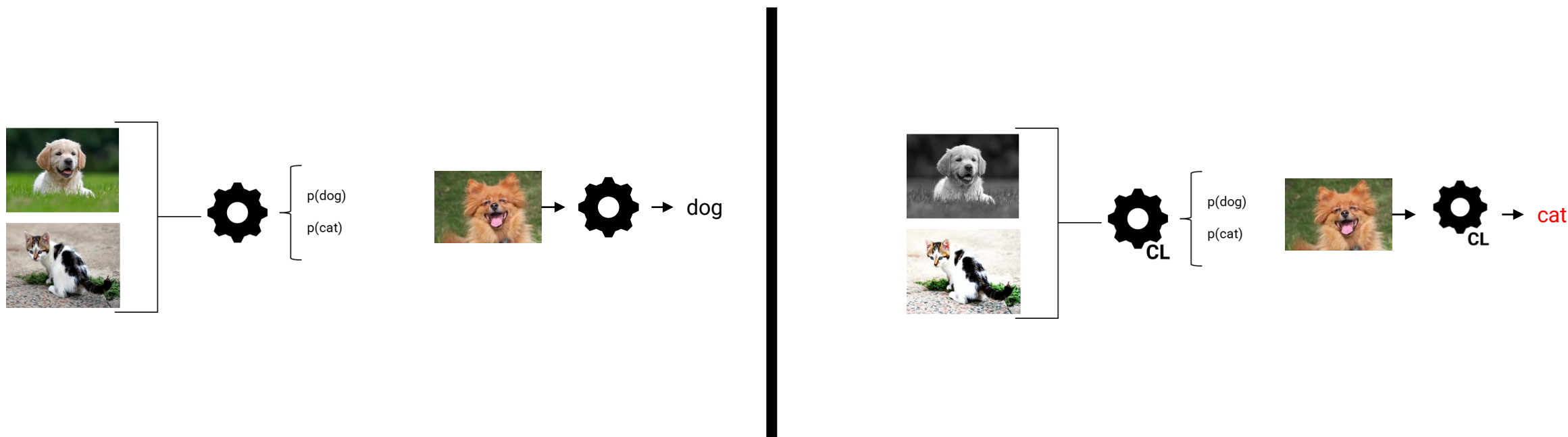
Embedding Initialization

In KGEs, the embeddings are the **model weights** (i.e., parameters). Therefore, in the CKGE setting, **new weights** are introduced in **each update**.



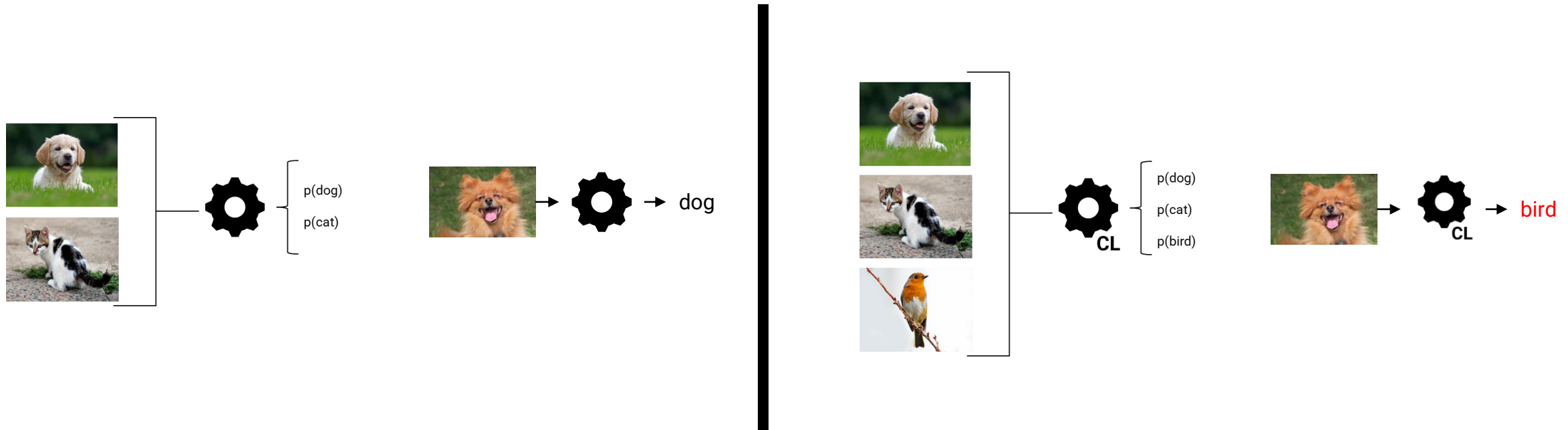
Embedding Initialization

In contrast to **other Continual Learning** scenarios, where there are **no new weights** or only a **small fraction** of them, in **CKGE** the proportion may be **significant**.



Embedding Initialization

In contrast to **other Continual Learning** scenarios, where there are **no new weights** or only a **small fraction** of them, in **CKGE** the proportion may be **significant**.



Embedding Initialization

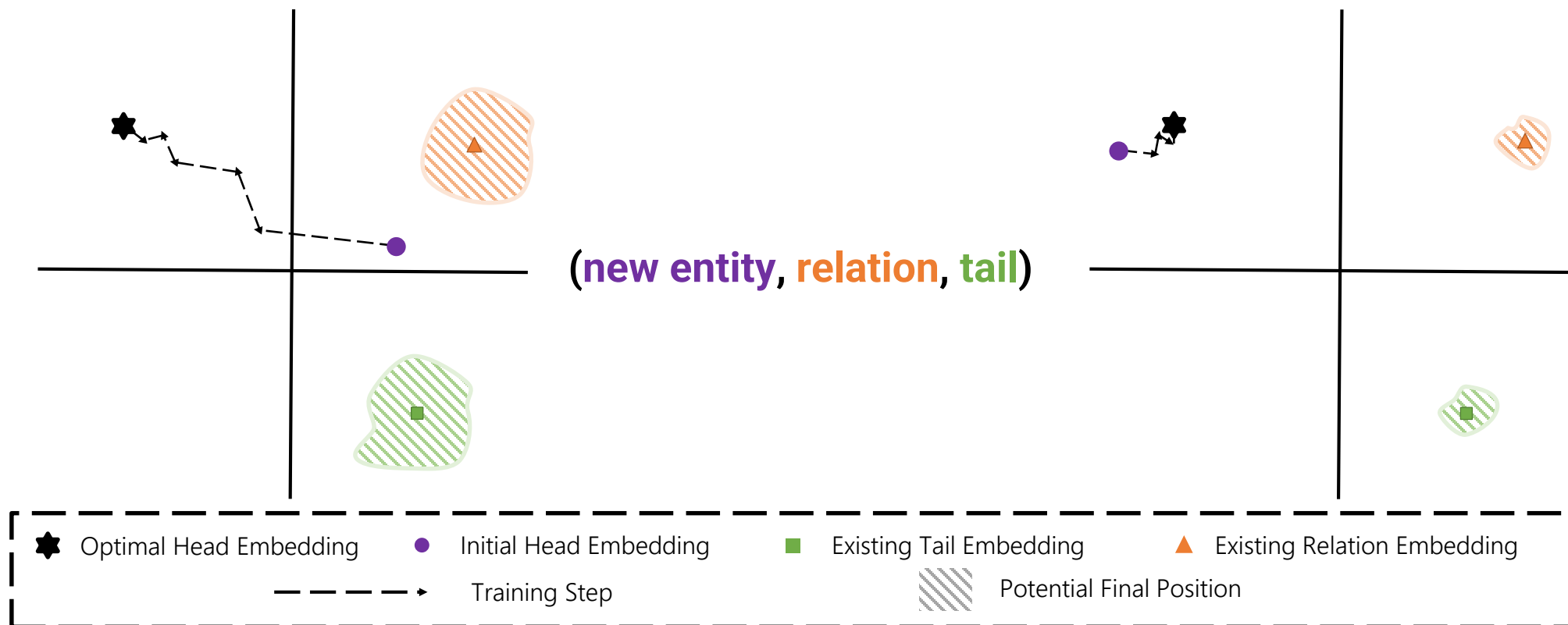
In contrast to **other Continual Learning** scenarios, where there are **no new weights** or only a **small fraction** of them, in **CKGE** the proportion may be **significant**.



From 40M entities in 2019 to 90M entities in 2021

Embedding Initialization: Random Initialization

New embeddings are typically **randomly initialized**. From this random position, their values get **adjusted during the training process**. This adjustment affects the other existing embeddings.



Embedding Initialization: Random Initialization

New embeddings are typically **randomly initialized**. From this random position, their values get **adjusted during the training process**. This adjustment affects the other existing embeddings:

- As the triples where new entities participate it are **not correctly predicted** at the start (i.e., high loss), all the elements in the triple are effected by **backpropagation**. This can result in unwanted changes.
- It takes time (i.e., **epochs**) to move the embedding from its **random initial** position to its **final value**.

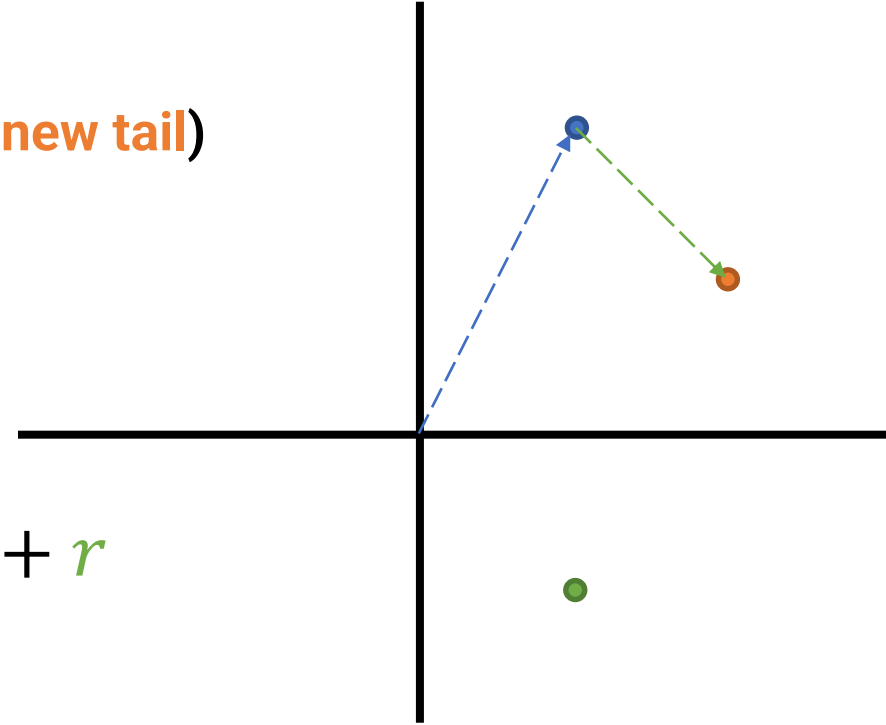
Embedding Initialization: Model Initialization

Some works [2, 3] have proposed initializing the embeddings with the **expected position** according to the **KGE model**.

(**head**, **relation**, **new tail**)

In the TransE model: $h + r = t$

$$t_{init} = h + r$$



[2] Y. Cui, Y. Wang, Z. Sun, W. Liu, Y. Jiang, K. Han, W. Hu, Lifelong embedding learning and transfer for growing knowledge graphs, in: AAAI, 2023.

[3] C. Wewer, F. Lemmerich, M. Cochez, Updating embeddings for dynamic knowledge graphs, arXiv preprint arXiv:2109.10896 (2021).

Embedding Initialization: Model Initialization

Some works [2, 3] have proposed initializing the embeddings with the **expected position** according to the **KGE model**.

This is done by **averaging** the expected position for the new entity for all the **triples** it **participates in**.

$$e_{init}^{model} = \frac{1}{|triples_e|} \sum_{t \in triples_e} f_{expected}(t)$$

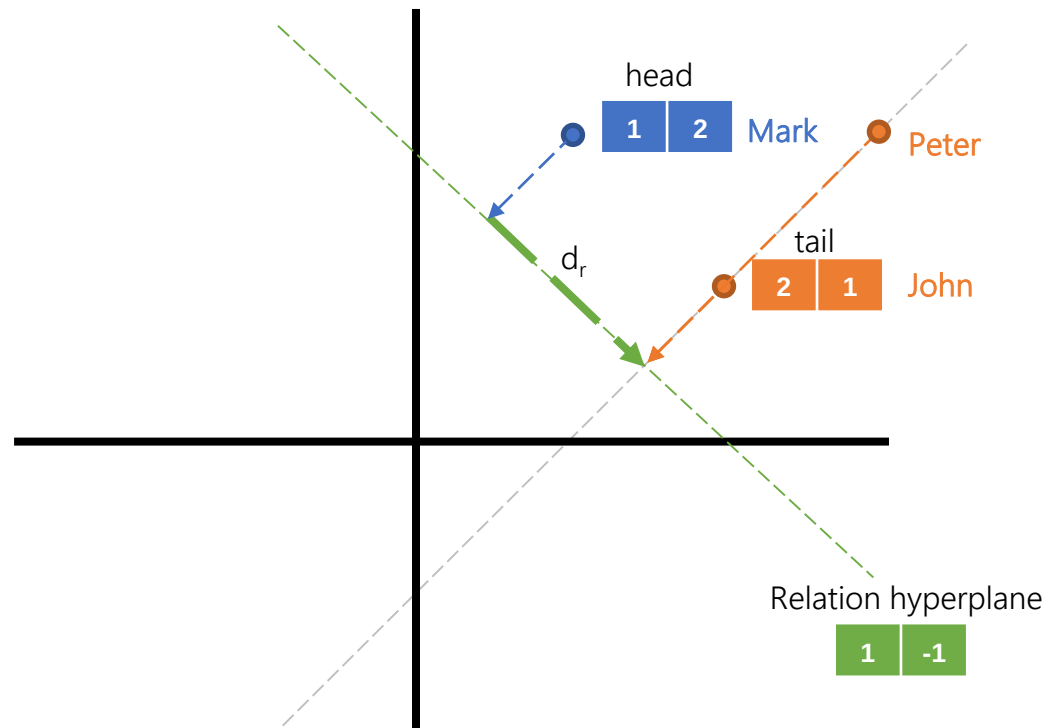
[2] Y. Cui, Y. Wang, Z. Sun, W. Liu, Y. Jiang, K. Han, W. Hu, Lifelong embedding learning and transfer for growing knowledge graphs, in: AAAI, 2023.

[3] C. Wewer, F. Lemmerich, M. Cochez, Updating embeddings for dynamic knowledge graphs, arXiv preprint arXiv:2109.10896 (2021).

Embedding Initialization: Model Initialization

However, this method has some **drawbacks**:

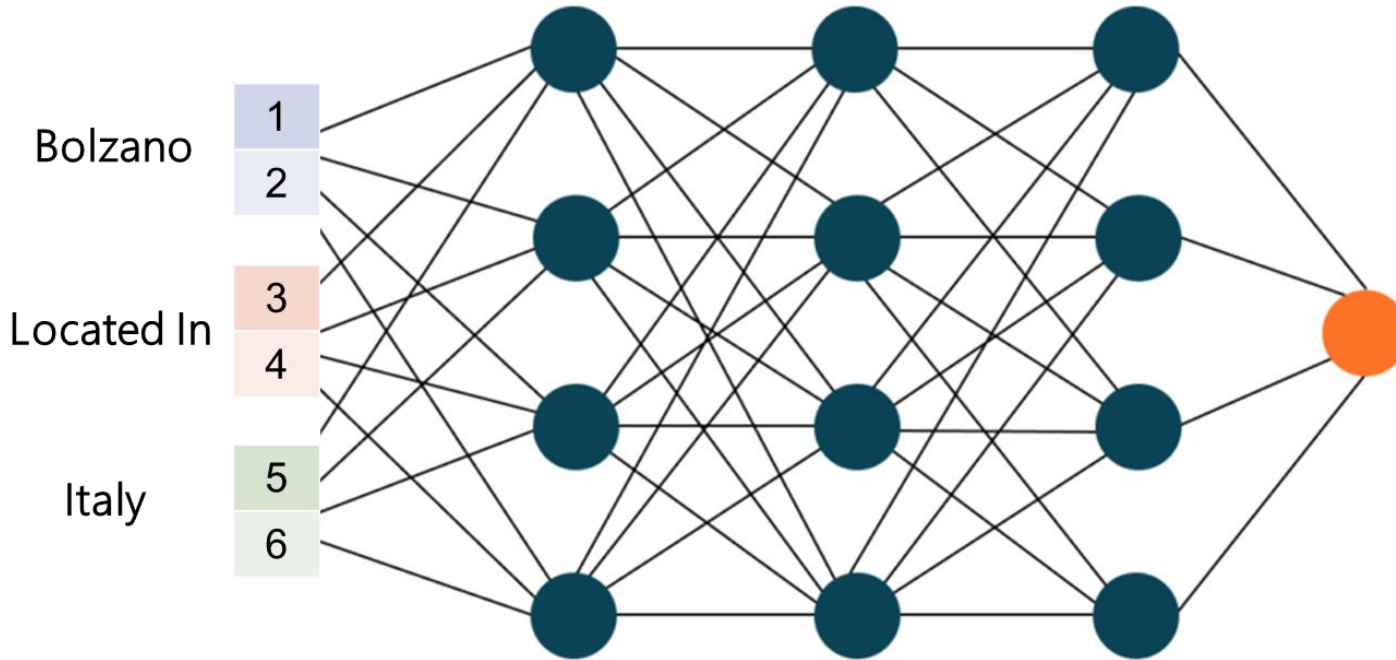
- For a triple to be directly **useful**, it can only contain **one new embedding**.
- The embedding function needs to be **invertible**.



Embedding Initialization: Model Initialization

However, this method has some **drawbacks**:

- For a triple to be directly **useful**, it can only contain **one new embedding**.
- The embedding function needs to be **invertible**.

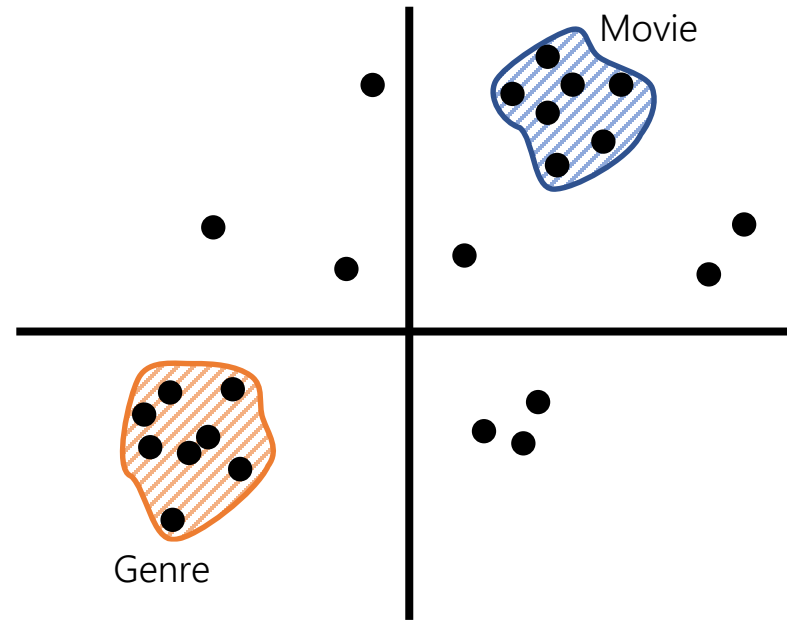


Non-Linear Activation functions: cannot be inverted.

Thousands of extra parameters: how to derived a closed-form analytical solution?

Embedding Initialization: Schema Initialization

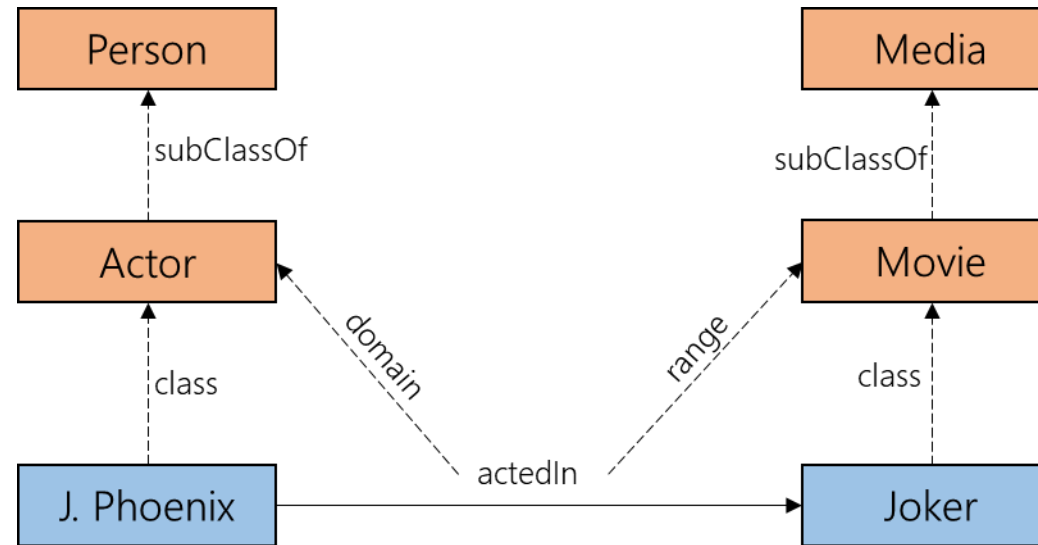
The **KG's schema**, can be used to provide an **informed initialization** that does not depend on the KGE model or the triples new embeddings appear in.



Embedding Initialization: Schema Initialization

The **KG's schema**, can be used to provide an **informed initialization** that does not depend on the KGE model or the triples new embeddings appear in:

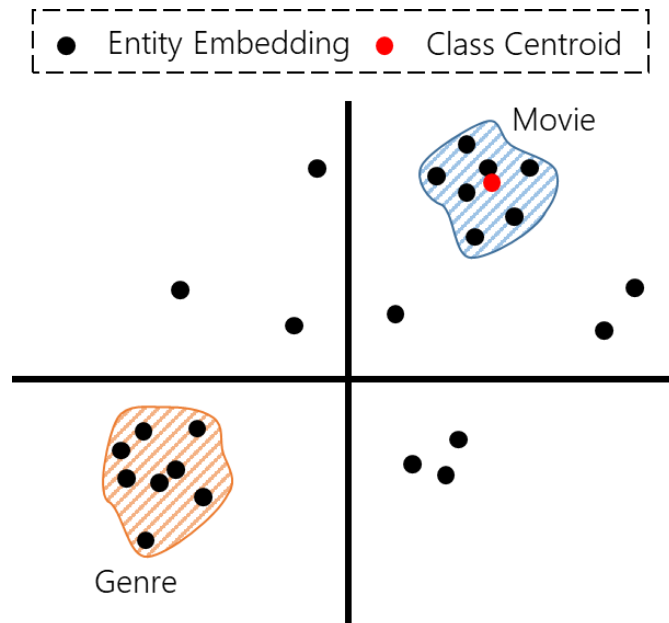
1. When a new entity arrives, we retrieve the **associated classes**.



Embedding Initialization: Schema Initialization

The **KG's schema**, can be used to provide an **informed initialization** that does not depend on the KGE model or the triples new embeddings appear in:

1. When a new entity arrives, we retrieve the **associated classes**.
2. We compute the **centroid** for all the existing embeddings that belong to each class.



Embedding Initialization: Schema Initialization

The **KG's schema**, can be used to provide an **informed initialization** that does not depend on the KGE model or the triples new embeddings appear in:

1. When a new entity arrives, we retrieve the **associated classes**.
2. We compute the **centroid** for all the existing embeddings that belong to each class.
3. We initialize the embedding by **averaging the centroids** of all the classes the entity belongs to.

$$e_{init}^{schema} = \frac{1}{|C_e|} \sum_{c \in C_e} centroid_c + random_noise$$

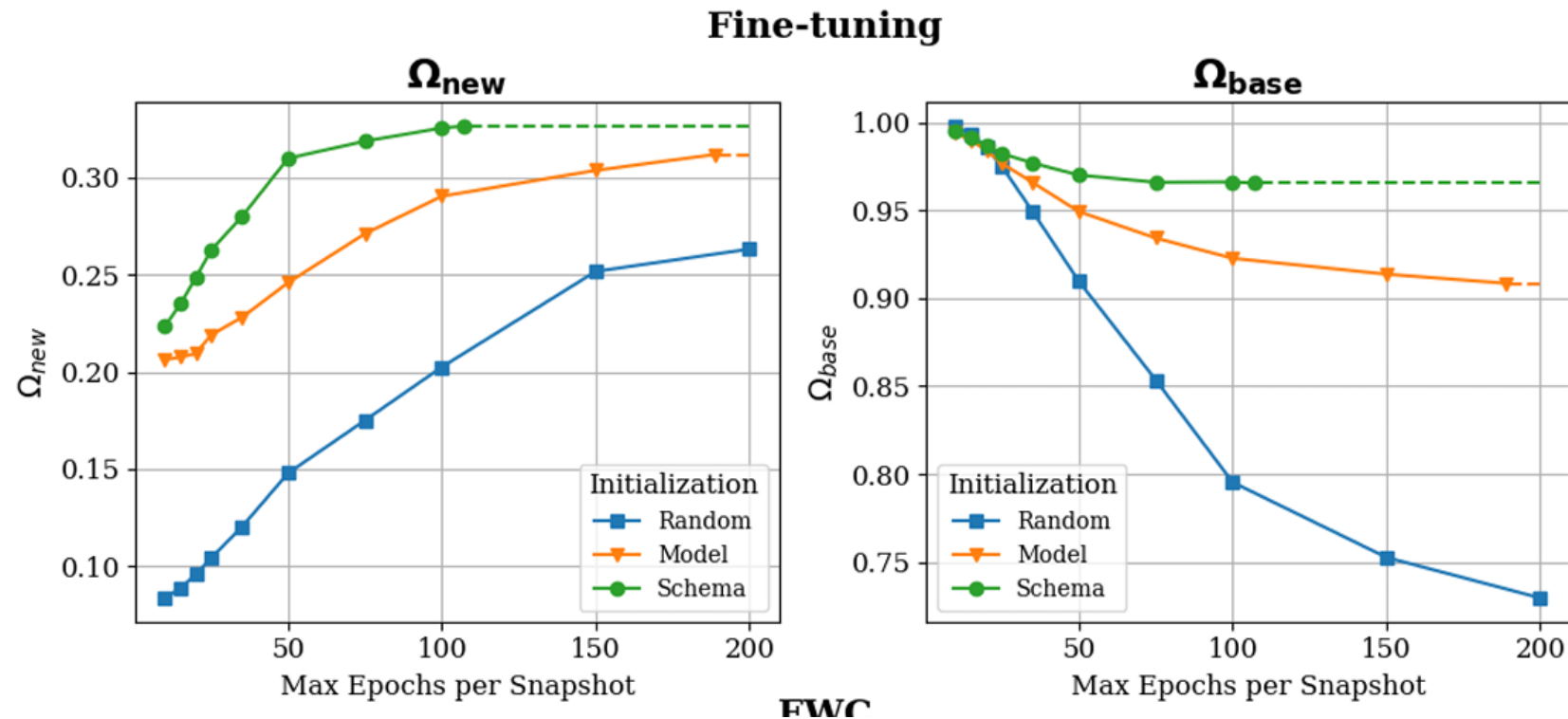
Embedding Initialization: Schema Initialization

This initialization improves **knowledge acquisition** and **retention**.

Method	Init.	FBinc-S				FBinc-M				FBinc-L			
		MRR	H@3	Ω_{base}	Ω_{new}	MRR	H@3	Ω_{base}	Ω_{new}	MRR	H@3	Ω_{base}	Ω_{new}
Retrain	–	0.302	0.354	–	–	0.302	0.352	–	–	0.300	0.351	–	–
FT	Rand.	0.204	0.249	0.779	0.321	0.193	0.232	0.735	0.254	0.187	0.220	0.723	0.284
	Model	0.246	0.297	0.871	0.532	0.258	0.306	0.914	0.302	0.237	0.276	0.825	0.301
	Schema	0.265	0.316	0.920	0.536	0.282	0.335	0.964	0.332	0.268	0.313	0.920	0.329
EWC	Rand.	0.232	0.276	0.838	0.284	0.232	0.268	0.809	0.284	0.259	0.301	0.922	0.310
	Model	0.264	0.312	0.912	0.465	0.274	0.323	0.948	0.273	0.265	0.310	0.905	0.358
	Schema	0.280	0.330	0.953	0.511	0.286	0.333	0.969	0.314	0.266	0.311	0.932	0.392
EMR	Rand.	0.202	0.244	0.771	0.311	0.196	0.232	0.746	0.239	0.186	0.217	0.715	0.276
	Model	0.249	0.296	0.872	0.494	0.264	0.311	0.930	0.287	0.238	0.276	0.832	0.306
	Schema	0.267	0.317	0.927	0.514	0.282	0.330	0.958	0.330	0.270	0.315	0.928	0.319
LKGE	Rand.	0.243	0.285	0.844	0.279	0.257	0.294	0.830	0.317	0.286	0.333	0.953	0.340
	Model	0.290	0.342	0.953	0.552	0.297	0.348	0.972	0.363	0.290	0.339	0.938	0.378
	Schema	0.293	0.343	0.964	0.513	0.298	0.346	0.978	0.331	0.299	0.350	0.962	0.382
incDE	Rand.	0.285	0.339	0.875	0.358	0.299	0.353	0.911	0.412	0.301	0.358	0.911	0.439
	Model	0.309	0.368	0.938	0.551	0.311	0.369	0.925	0.463	0.308	0.365	0.918	0.424
	Schema	0.315	0.376	0.960	0.539	0.316	0.378	0.942	0.491	0.308	0.368	0.926	0.445

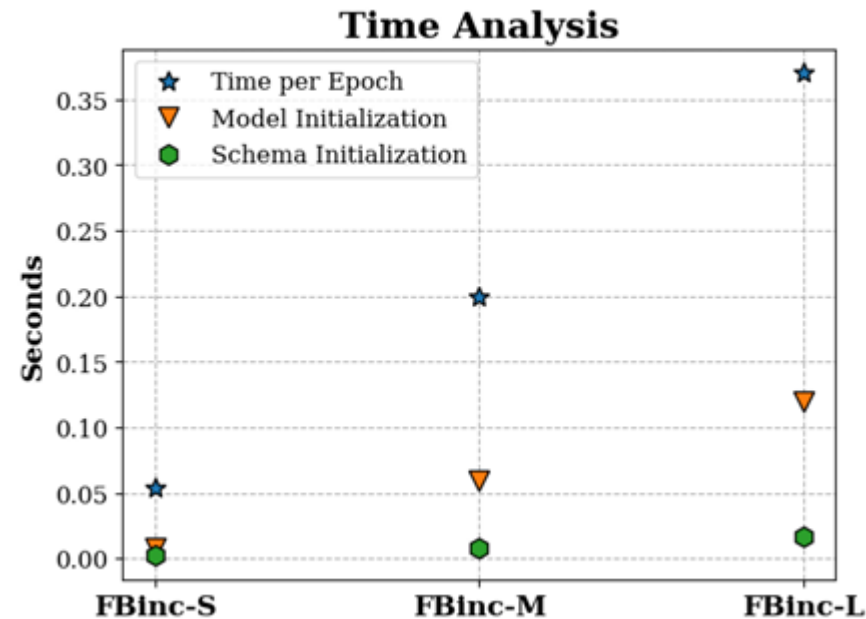
Embedding Initialization: Schema Initialization

At the same time, it also **decreases training time**:



Embedding Initialization: Schema Initialization

At the same time, it also **decreases training time**.



Embedding Initialization: Schema Initialization

It is also applicable to **different KGE models**.

Family	Model	Init.	FBinc-S			FBinc-M			FBinc-L		
			H@3	Ω_{base}	Ω_{new}	H@3	Ω_{base}	Ω_{new}	H@3	Ω_{base}	Ω_{new}
Translational	TransH	Random	0.272	0.809	0.471	0.314	0.909	0.236	0.268	0.809	0.260
	TransH	Schema	0.322	0.901	0.516	0.343	0.944	0.344	0.333	0.918	0.328
	RotatE	Random	0.212	0.567	0.231	0.266	0.633	0.203	0.300	0.714	0.359
	RotatE	Schema	0.351	0.894	0.279	0.282	0.796	0.246	0.310	0.735	0.431
	TransR	Random	0.152	0.719	0.254	0.139	0.645	0.198	0.086	0.438	0.232
	TransR	Schema	0.213	0.901	0.245	0.217	0.936	0.193	0.165	0.717	0.246
Semantic	DistMult	Random	0.268	0.782	0.257	0.317	0.921	0.146	0.305	0.945	0.148
	DistMult	Schema	0.330	0.942	0.274	0.346	0.973	0.232	0.331	0.975	0.223
	HolE	Random	0.252	0.899	0.127	0.263	0.957	0.094	0.235	0.938	0.110
	HolE	Schema	0.254	0.915	0.317	0.303	0.995	0.280	0.277	0.971	0.241
	ProjE	Random	0.351	0.994	0.176	0.318	0.964	0.181	0.271	0.919	0.134
	ProjE	Schema	0.352	0.994	0.238	0.326	0.966	0.247	0.281	0.919	0.190

Ch 3: CKGE

Experimentation

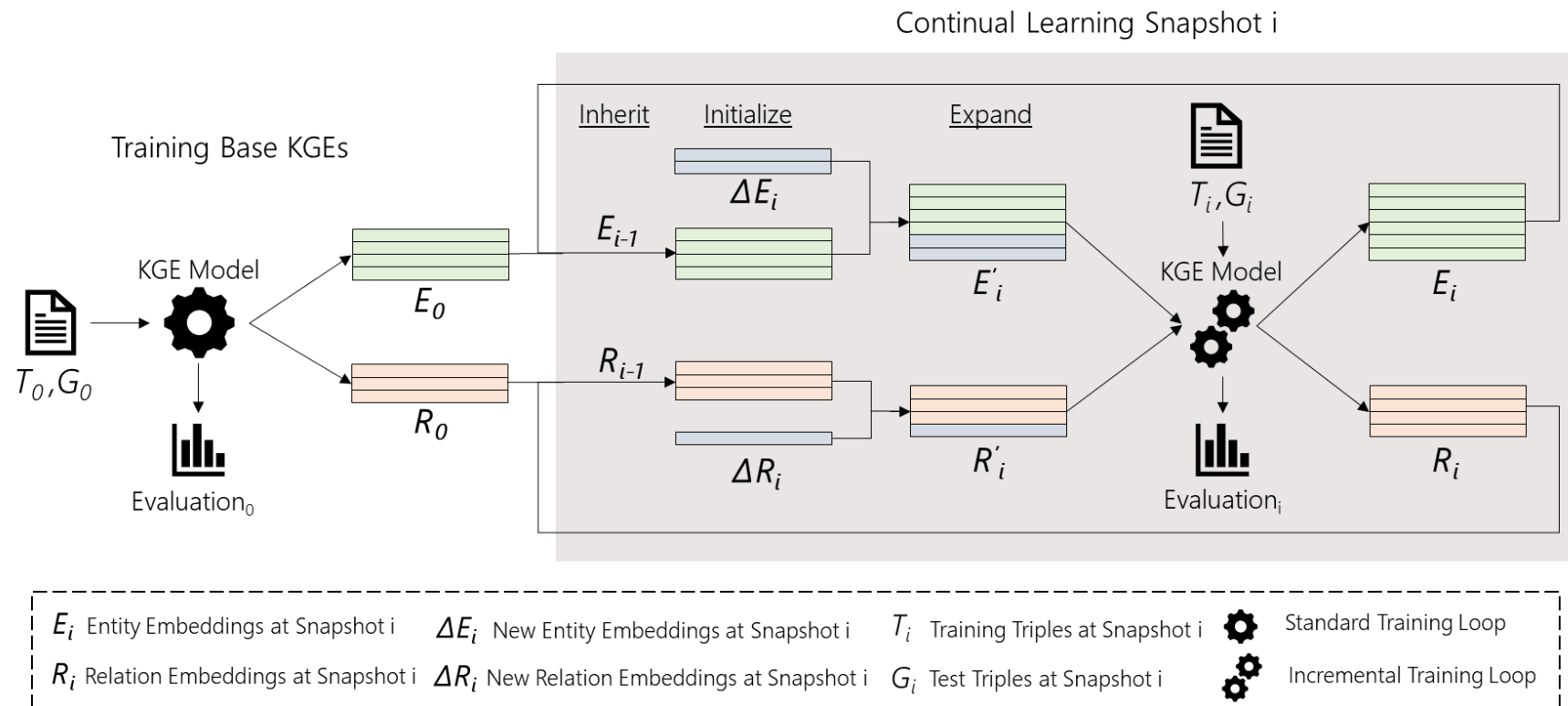
CKGE Experimentation:

CKGE is a **necessary** and **rapidly growing field**. However, its development has **not** been **aligned** with common **KGE frameworks and libraries** (e.g., PyKEEN):

- CKGE methods are applied within their **own frameworks**, without using different **KGE models** (i.e., CKGE methods rely on TransE) or benefiting from more advanced **KGE training techniques** (e.g., negative sampling).
- At the same time, KGE libraries do **not support evolving KGEs** (e.g., metrics, evolution...).

CKGE Experimentation:

The **PyKEEN** library has been extended [4] to accommodate the **CKGE framework**.



[4] G. Pons, B. Bilalli, and A. Queralet. "A Framework for Continual Knowledge Graph Embedding" in *Companion Proceedings of the ACM Web Conference*. 2026

CKGE Experimentation:

The **PyKEEN** library has been extended to accommodate the **CKGE framework**:

- Appropriately expanding the KGE models to accommodate **entity** and **relation** growth.
- Implementation of **a Continual Learning training loop**.
- CKGE **evaluation** and **metrics**.
- Support of **custom initialization methods**.

CKGE Experimentation:

FUTURE WORK:

Currently, only **fine-tuning** is supported. Therefore, different SoTA CKGE methods should be implemented inside the **PyKEEN framework**.

Questions?